Quantum Checkers User Guide

Create your account, join your classroom and start solving quantum puzzles.

Game: https://quantum-checker.onrender.com/

1. Before you register

Ask your administrator or teacher which classroom to select, for example **Fall 2026**. They must add the classroom before it can appear in registration.

Have your student ID ready. It must start with an uppercase **D** followed by digits only, for example `D00123456`. Keep any zeros after the D. Do not invent an ID or use someone else's.

Accepted format	Not accepted
`D00123456`, `D123456`	`d123456`, `123456`, `D12A456`, `D12-456`, `D 123456`

2. Create an account

1. Open the game, click **Log in**, then **Create an account**. You can also open https://quantum-checker.onrender.com/signup.
2. Choose a **Username**. Remember it: this is what you will use to log in.
3. Enter your email if you wish. Email is optional.
4. Enter your **Student ID**, such as `D00123456`.
5. Choose your class, such as **Fall 2026**, from **Classroom (optional)**. Although the website allows Unassigned, choose the class your teacher gives you.
6. Choose a password with at least 8 characters. Avoid common passwords, an all-number password or a password too similar to your username.
7. Click **Sign up**. After successful registration, the login page opens.

> If your classroom is missing, ask your administrator to add it, then refresh the registration page. Each username and student ID can be used by only one account. If your ID is already registered, use your existing account or ask your administrator for help.

3. Log in

1. Enter your **Username** and password.

2. Click **Log in**.
3. On the main menu, click **Start Game** and choose a level.

Your student ID and optional email are separate from your login username. They only work in the Username box if you originally chose one of them as your username.

4. Solve a puzzle

The **Current** board shows the present state. The **Target** board shows what you need to reach. Use the enabled gate buttons to change Current until it matches Target.

Circle appearance	Value
Black	1
White	0
Outline	0.5
Dashed target	Any value is accepted at this position.

1. Read the level's instructions.
2. For **X**, **Z** or **H**, select **Left qubit** or **Right qubit** under **Apply a gate to**.
3. Click an enabled gate button. The board and move count update.
4. **CZ** always acts on both qubits, regardless of which side is selected.
5. When the target is reached, you see **Puzzle solved!**. Click **Choose another level** to continue.

Disabled buttons indicate gates that are not available in this level. If a level lists **Extra targets**, those conditions must also be met.

Practice with the first example

If the first sample level is still named **Flip the left qubit**, select **Left qubit** and apply **X** once. This should match the target and solve the puzzle. Your administrator may change or replace the example levels.

5. Restart or return later

Click **Reset puzzle** to start the current level again. This clears that level's moves and solved state. Click **All levels** to return to the level list.

Your moves are saved to your account as you play. Returning to a level normally resumes your latest attempt. If your administrator changes the puzzle's starting state, targets or available gates, reopening it starts the updated puzzle.

Use **Log out** on the main menu when you finish, especially on a shared computer.

6. Get help

Problem	What to do
Registration rejects my ID.	Use uppercase D followed by digits only. Remove spaces, hyphens and other letters. Keep your ID's zeros.
My class is missing.	Ask your administrator to add it and refresh the registration page.
My username or ID already exists.	Try your existing account. Ask your administrator for help if you cannot log in.
I forgot my password.	Ask your administrator to reset it. There is no automatic email reset.
I selected the wrong class or entered the wrong ID.	Ask your administrator to correct your Student profile. Do not register another account.
The website loads slowly on my first visit.	The free website may need time to wake up. Wait and retry.
The level says it was edited.	Refresh the page to load the updated puzzle.
